



Fig. LOGO

Team **SoloSphere**

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PROJECT PROPOSAL

Samadhan

“Bringing Solutions Closer to Home”

Connecting Local professionals to Household Service
needs

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1. Summary

In Bangladesh, **individuals frequently encounter various household and technical issues, such as plumbing, electrical problems, furniture repair, and appliance maintenance. Locating and connecting with local service providers, also known as service professionals, for these services can be a challenging and time-consuming process.** There's a need for a convenient, efficient, and user-friendly platform that allows users to post their service requirements, negotiate with local service professionals, and receive timely assistance for their home-related issues.

The system can bring the following benefits to our daily social life:

1. Convenience: Somadhan simplifies the process of finding and connecting with local service professionals, making it a convenient one-stop platform for home-related service needs.
2. Efficiency: The app streamlines communication, negotiation, and scheduling, saving time for both users and service professionals.
3. Trust Building: The rating and review system and penalty mechanism foster trust within the community, ensuring quality service.
4. Economic Empowerment: The app empowers local service professionals by providing them with a platform to expand their customer base and grow their businesses.

2. Introduction

Name	Contribution
Abu Saleh Md. Boni Momin ID: 011193076	Literally Everything!!! Trust me! From the beginning of requirement analysis, system and architectural Design, FrontEnd development using React-native, BackEnd development using Nest.JS, and database manipulation on MySQL; the one man Abu Saleh... handled it ALL.

3. Tech Stack & Architecture

Tech Stack

Frontend: React Native

Backend: Nest.js

Database: MySQL

Architectural Pattern: 4-Tier Layered Architecture

In your **4-tier layered architecture**, each layer plays a specific role in the functioning of the system:

1. Presentation Layer:

The Presentation Layer is the user interface of your mobile application built with React Native. It handles user interaction, input validation, and the rendering of views.

2. Business Layer:

The Business Layer, often referred to as the Application Layer, contains the application's business logic and rules.

3. Persistence Layer:

The Persistence Layer is responsible for handling data storage and retrieval. It interacts with the database (MySQL) to store user profiles, service requests, chat messages, and other data.

4. Database Layer:

The Database Layer includes the MySQL database, where data is stored in structured tables. It is responsible for data storage, retrieval, and management.

4. Features/Objectives

The “Samadhan” platform is consisted with the following features.

User Registration and Profiles:

- Users can create accounts with their details, location, and contact information.
- Service professionals can create profiles with their expertise, service categories, and contact details.

Issue Posting:

- Users can post service requests by capturing images and videos of the problem and providing a detailed description.

Location-Based Service Discovery:

- Users can view nearby service professionals based on their current location.

Chat and Negotiation:

- Users and service professionals can initiate a chat to discuss service details and negotiate prices and service schedules.

Service Scheduling:

- Once a service request is confirmed, both parties can agree on a service time within the chat.

Invisibility on Confirmation:

- After a service professional confirms a job, the service request becomes invisible to other service professionals.

Multi-Platform Support:

- The app is accessible on both iOS and Android platforms.

Penalty System:

- Service professionals are penalized if they don't arrive at the agreed service time, and the request becomes visible to other professionals.

Ratings and Reviews:

- Users and service professionals can rate and review each other, building trust within the community.

Notification System:

- Users receive notifications for chat messages, service confirmations, and service completions.

Service History:

- Maintain a history of past service requests and completed jobs for both users and service professionals.

5. Future Plan

Advanced Search Filters:

Enhance the search functionality with filters based on service type, pricing, ratings, and more.

Video Calls:

Implement video call capabilities to allow for remote diagnosis of issues and consultation.

Service Categories Expansion:

Include more service categories such as gardening, cleaning, and pest control.

Subscription Model:

Introduce a subscription-based premium service with added benefits for users and service professionals.